

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1_AT3 — VISUALIZATION DESIGN EXERCISE

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO1 – Precision (BL2)	Assessment:	Assessment Tool 3 – Visualization Design
Weightage:	40% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO1 Statement:	<i>Understand the fundamental concepts, principles, and techniques of data visualization.(BL2)</i>		
Student Name:	T. Nivya Sri	Reg. No.:	192424131
Section / Batch:	2024	Date:	28-07-26

Code of Conduct

I Nivya Sri. T (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This assessment contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Annexure A – VISUALIZATION DESIGN

ASSESSMENT RUBRIC

AT3 — Visualization Design Exercise

CO1 · BT3: Apply ·

This rubric is used to assess student responses to the AT3 Visualization Design Exercise problems, in which students design and justify a complete visualization (chart type, aesthetic mappings, scales, coordinate systems, and/or color scheme) for a given dataset or design brief.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of chart types, aesthetic mappings, scales, coordinate systems, and color scales, and correctly identifies which apply to the given design scenario.	Good understanding of the relevant design concepts, with only minor gaps in selecting appropriate chart types, aesthetic mappings, or scales.	Basic understanding of design concepts, but with some misconceptions about appropriate chart types, aesthetics, or scales for the scenario.	Poor understanding of core design concepts; proposes chart types or mappings that are fundamentally unsuited to the data.
Explanation	25%	Provides detailed, well-reasoned justification for every design decision, explicitly connecting each choice to the properties of the data and the scenario.	Provides clear justification for design decisions, with adequate reasoning for most sub-questions.	Justification is limited or superficial; design choices are stated without sufficient reasoning.	Little or no justification provided; design choices appear arbitrary or unexplained.
Application	20%	Effectively applies visualization design principles to produce a complete, appropriate, and well-reasoned design solution for the specific scenario.	Applies design principles to produce an appropriate solution, with only minor errors or missed refinements.	Limited application of design principles; the proposed design partially addresses the scenario but has notable weaknesses.	Fails to apply appropriate design principles; the proposed design does not suit the scenario or data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Organization	15%	The design proposal is clearly structured, addressing each sub-question (a, b, c) in a logical sequence with coherent flow throughout.	The proposal is organized and addresses each sub-question, with only minor issues in flow or clarity.	Basic structure; sub-questions are addressed but the proposal lacks clarity or a logical sequence.	Poorly organized; the proposal is incoherent, and sub-questions are left unaddressed or answered out of order.
Accuracy	10%	All design choices — chart type, aesthetic mapping, scale, coordinate system, and/or color scheme — are completely accurate and well-suited to the data.	Mostly accurate design choices, with only minor omissions or a small error in one part of the proposal.	Partially correct; some design elements are missing, mismatched to the data, or incorrect.	Design choices are largely incorrect or inappropriate, with major gaps in matching the visualization to the data.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO1 AT3: VISUALIZATION DESIGN EXERCISE

Scenario: A biotech company measures the concentration of a compound in a solution, with values doubling every trial across 10 trials, ranging from 1 to 1024 units.

a) Design a chart to display this data. Which axis type (linear or logarithmic) would you choose, and why?

A line chart is the most suitable way to display the compound concentration measured across 10 trials because the data is sequential and increases exponentially. A logarithmic y-axis is the correct choice since the concentration values double each trial, ranging from 1 to 1024 units. On a linear axis, the early values such as 1, 2, 4, and 8 would appear compressed near the bottom of the chart, while the later values like 512 and 1024 would dominate the visual space. This distorts the true pattern of exponential growth. A log axis spreads multiplicative changes evenly, making each doubling appear as an equal vertical step and revealing the consistent exponential trend clearly.

b) Describe how the spacing between data points would differ between a linear and a log axis for this dataset.

The spacing between data points differs significantly between linear and logarithmic axes. On a linear axis, spacing increases dramatically as values grow because equal physical distances represent equal absolute differences. The jump from 1 to 2 is visually tiny, while the jump from 512 to 1024 is extremely large. This causes the early part of the chart to look flat and the later part to rise sharply. On a log axis, spacing becomes uniform because equal ratios (such as doubling) are represented by equal distances. The sequence 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024 appears evenly spaced, accurately reflecting the exponential nature of the dataset.

c) What coordinate system (Cartesian or polar) is most appropriate here, and why?

A Cartesian coordinate system is the most appropriate choice for this visualization. The data progresses across ordered trials, and Cartesian coordinates are ideal for showing trends, slopes, and growth patterns over time. Polar coordinates are better suited for cyclical data or multi-variable comparisons, and using them here would distort the interpretation of exponential progression. A Cartesian line chart with a logarithmic y-axis therefore provides the clearest and most accurate representation of the compound concentration across trials.

VIZUALIZATION:



Scenario: You want to draw attention to only the single worst-performing product (out of 40) in a company's annual product performance report.

a) Design a color-highlighting scheme for this chart. What color would you use for the highlighted product, and what color for the rest?

To draw attention to the single worst-performing product out of forty, the most effective color-highlighting scheme is one where the highlighted product uses a strong, attention-grabbing color such as red, while all other products are shown in a neutral, low-salience color such as light grey. Red immediately signals "problem," "risk," or "requires attention," and the contrast against grey ensures the viewer's eye is drawn directly to the worst performer without distraction. This approach creates a clear visual hierarchy: one item stands out, and the rest fade into the background.

b) Explain why using this highlighting technique on more than 3–4 products would reduce its effectiveness.

This technique loses effectiveness if applied to more than three or four products because the visual contrast becomes diluted. When too many items are highlighted, the viewer's attention becomes scattered, and the chart loses its focal point. Instead of guiding the viewer to a single critical insight, the chart becomes visually noisy, making it harder to interpret and reducing the impact of the highlight. Highlighting works only when it is rare; once it becomes common, it stops functioning as a signal.

c) Would you use this same highlight color elsewhere in your report for unrelated purposes? Justify your answer regarding consistency.

The highlight color should not be reused elsewhere in the report for unrelated purposes. Red, once established as the “critical issue” color, must remain consistent in meaning throughout the report. Using it for unrelated elements—such as titles, decorative accents, or neutral categories—would confuse the viewer and weaken its communicative power. Consistency in color semantics is essential: a color used to indicate poor performance should be reserved exclusively for that purpose so that readers immediately understand its meaning wherever it appears.

VISUALIZATION:

